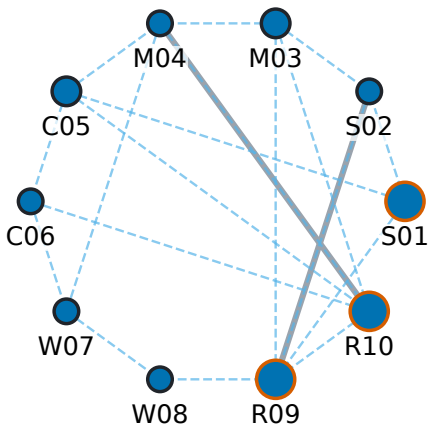


Functional ThermoHITL operator dashboard — populated ledger replay, step 11

Network and autonomy



Thermodynamic state

- energy 0.230
- entropy 0.380 · anomaly 5.11σ
- disagreement 0.015
- consensus confidence 0.942

Alert queue

- 1 queued incident(s)
- reason severity
- predicted collapse 4 periods
- view payload hashes 3

Latest bounded intervention

- event operator_result
- tool authorize_emergency_route
- status human_intervention_applied
- autonomy level 4

Operator workload

- workload 0.646
- active interventions 2
- fatigue 0.143
- minutes 24.0

Execution-time information boundary

The simulated operator and dashboard consume the same schema-validated, hashed payload. Private observations, RNG state, future disruptions, and counterfactual outcomes are excluded.

Replay controls: play · pause · step · rewind · jump to alert · compare branch · export SVG